



3?

5?

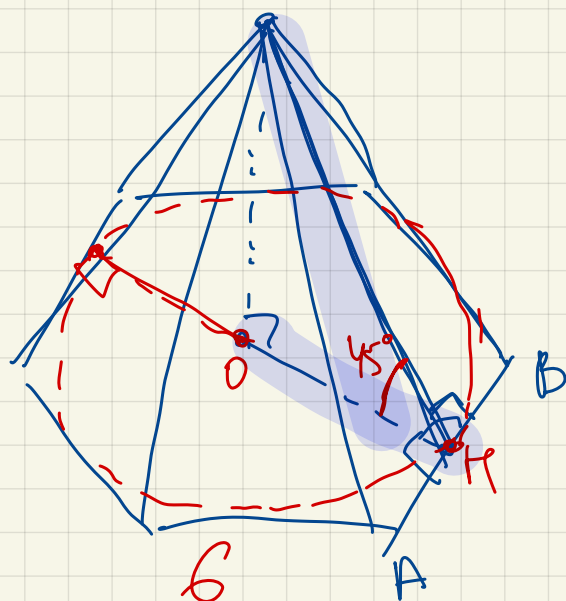
$$(x-2)(6-x) \leq \frac{x+3}{2}$$

$$2(6x - x^2 - 12 + 2x) \leq x + 3$$

$$12x - 2x^2 - 24 + 4x \leq x + 3$$

$$-2x^2 + \underbrace{16x}_{15x} - 27 \leq x + 3$$

$$\S \quad -2x^2 + \underline{\underline{15x}} - 27 \leq 0$$



$$S = p \cdot r$$

$$r = \frac{S}{p} =$$

$$S' = \frac{3a^2\sqrt{3}}{2}$$

$$p = \frac{6a}{2} = 3a$$

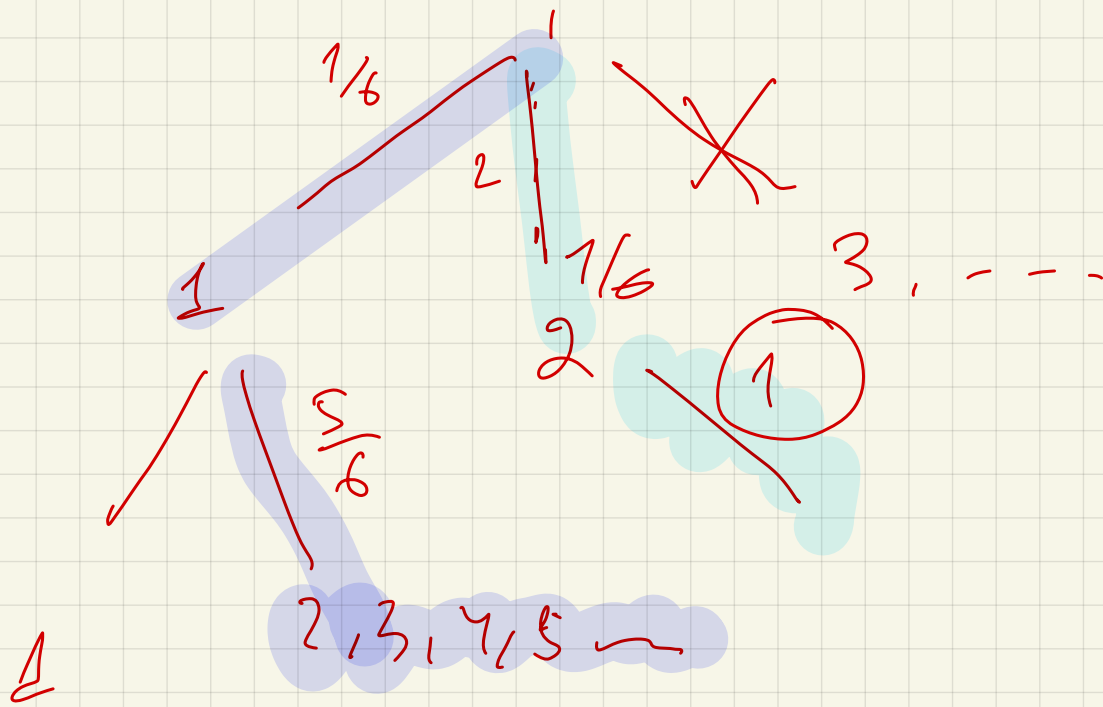
$$r = \frac{\frac{3a^2\sqrt{3}}{2} \cdot 3}{3a} = \frac{a\sqrt{3}}{2}$$

$$\frac{SO}{OH} = \tan 45^\circ = 1$$

$$SO = OH = \frac{a\sqrt{3}}{2}$$

$$V = \frac{1}{3} \cdot \frac{3a^2\sqrt{3}}{2} \cdot \frac{a\sqrt{3}}{2} = a^3 \cdot \frac{3}{4} =$$

$$\frac{6^3 \cdot 3}{4} = \frac{216 \cdot 3}{4} = \underline{\underline{162}}$$



Δ

> 2

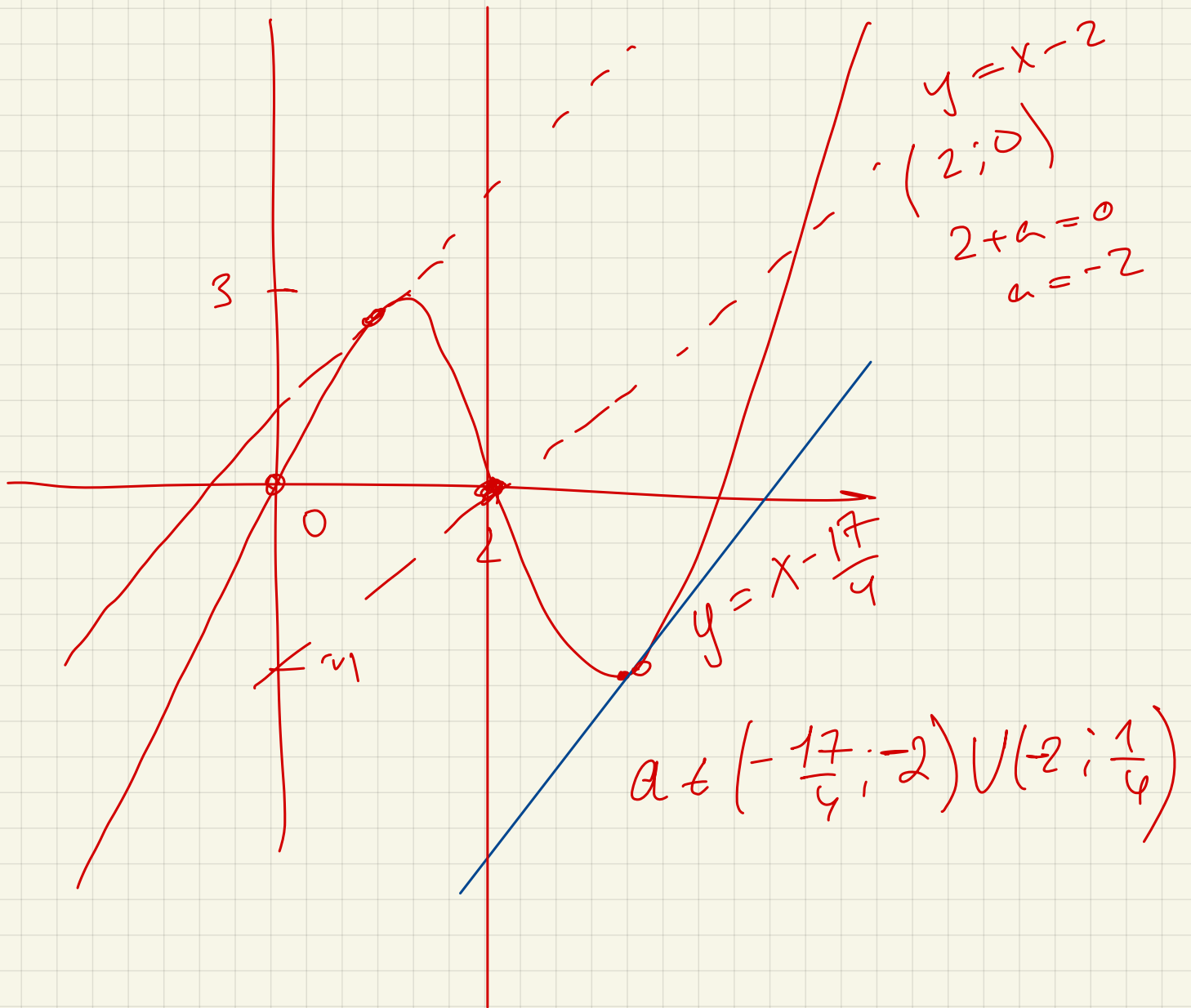
$$P_1 = \frac{1}{6} \cdot \frac{5}{6}$$

$$P_2 = \frac{1}{6} \cdot 1$$

$$P = P_1 + P_2 =$$

$$\frac{5}{36} + \frac{1}{6} = \frac{11}{36}$$

0,306



$$y = x^2 - 6x + 8$$

$$y = x + a$$

$$x + a = x^2 - 6x + 8$$

$$x^2 - 7x + 8 - a = 0$$

$$\Delta = 0$$

$$49 - 4(8 - a) = 0$$

$$49 - 32 + 4a = 0$$

$$17 + 4a = 0$$

$$a = -\frac{17}{4}$$

$$y = -x^2 + 2x$$

$$a = \frac{1}{4}$$

$$\begin{cases} |x-a| + |y| = 2 \\ y = \sqrt{4-x^2} \end{cases}$$

2 per

$$a = 1$$

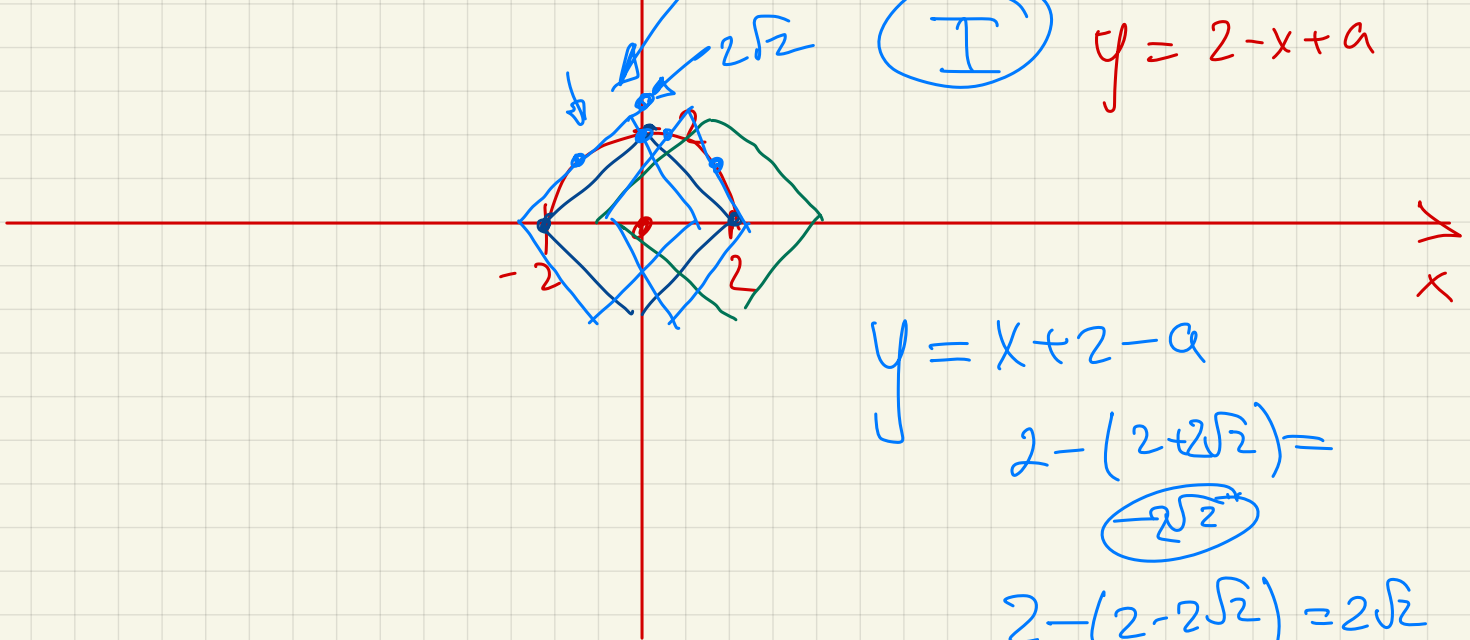
(II)

$$-x + a + y = 2$$

$$y = 2 + x - a$$

(I)

$$y = 2 - x + a$$



$$y = x + 2 - a$$

$$2 - (2 + 2\sqrt{2}) = -2\sqrt{2}$$

$$2 - (2 - 2\sqrt{2}) = 2\sqrt{2}$$

$$4 - x^2 \geq 0 \quad x^2 \leq 4$$

$$-2 \leq x \leq 2$$

$$y \geq 0$$

$$y^2 = 4 - x^2$$

$$\begin{aligned} &(-x + (2+a))^2 \\ &(-x)^2 + (2+a)^2 - 2 \cdot x(2+a) \end{aligned}$$

$$y^2 + x^2 = 2^2$$

$$(x + 2 - a)^2 + x^2 = 4$$

$$x^2 + (2-a)^2 + 2 \cdot x(2-a) + x^2 = 4$$

$$2x^2 + 2x(2-a) + 4 + a^2 - 4a - 4 = 0$$

$$2x^2 + 2x(2-a) + a^2 - 4a = 0$$

$$\Delta = 0$$

$$(2(2-a))^2 - 4(a^2 - 4a) \cdot 2 = 0$$

$$4(4 - 2a \cdot 2 + a^2) - 8a^2 + 32a = 0$$

$$-4a^2 + 16a + 16 = 0$$

$$-a^2 + 4a + 4 = 0$$

$$\frac{-4 \pm \sqrt{16 \pm 16}}{-2}$$

$$\frac{-4 \pm 4\sqrt{2}}{-2}$$

$$2 \pm 2\sqrt{2}$$

$$a_1 = 2 - 2\sqrt{2}$$

$$(-x + 2 + a)^2 + x^2 = 4$$

$$(-x + 2 + a)(-x + 2 + a) = x^2 - 2x - \cancel{a} - 2x + 4 + \underline{2a} - \underline{ax} + \underline{2a} + \underline{a^2}$$

$$x^2 - 4x - 2xa + 4a + a^2 + 4$$

$$2x^2 - x(4 + 2a) + a^2 + 4a = 0$$

$$\Delta = (4 + 2a)^2 - 4 \cdot 2(a^2 + 4a) = 0$$

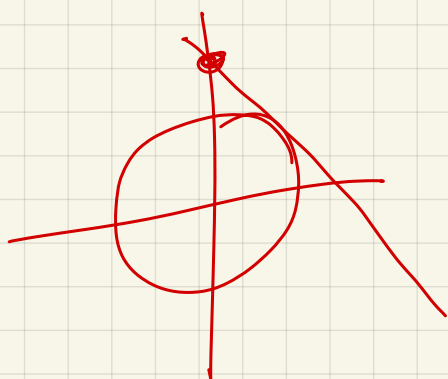
$$16 + 4a^2 + 16a - 8a^2 - 32a = 0$$

$$-4a^2 - 16a + 16 = 0$$

$$-a^2 - 4a + 4 = 0$$

$$\frac{4 \pm \sqrt{16 + 16}}{-2} = -2 \pm 2\sqrt{2}$$

$$\begin{aligned} & -2 - 2\sqrt{2} \\ & -2 + 2\sqrt{2} \end{aligned}$$



$$-x + 2 + a$$

$$a = -2 + 2\sqrt{2}$$

$$a = \{2 - 2\sqrt{2}; -2 + 2\sqrt{2}\}$$

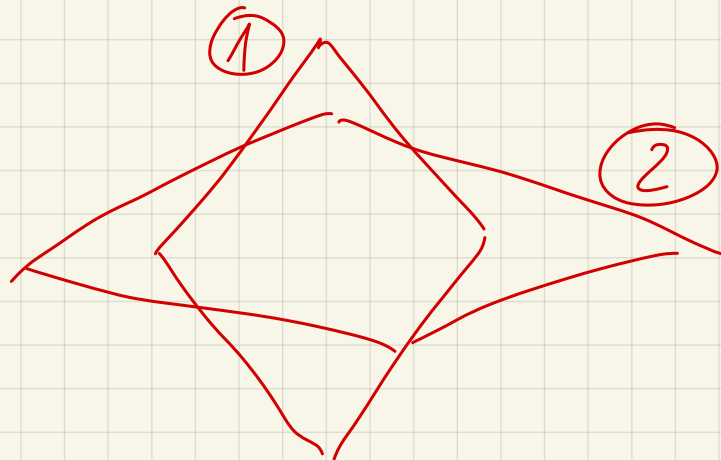
$$|3x| + |4y| = 12a$$

$$x^2 + y^2 - 10y = 0$$

2 per

$$|x| + |y| = 1$$

$$a = 1$$



$$a = 1$$

$$|3x| + |4y| = 12$$

$$|x| + |y| = 1$$

$$\textcircled{1} \quad 3x + 4y = 12$$

$$y = \frac{12 - 3x}{4}$$

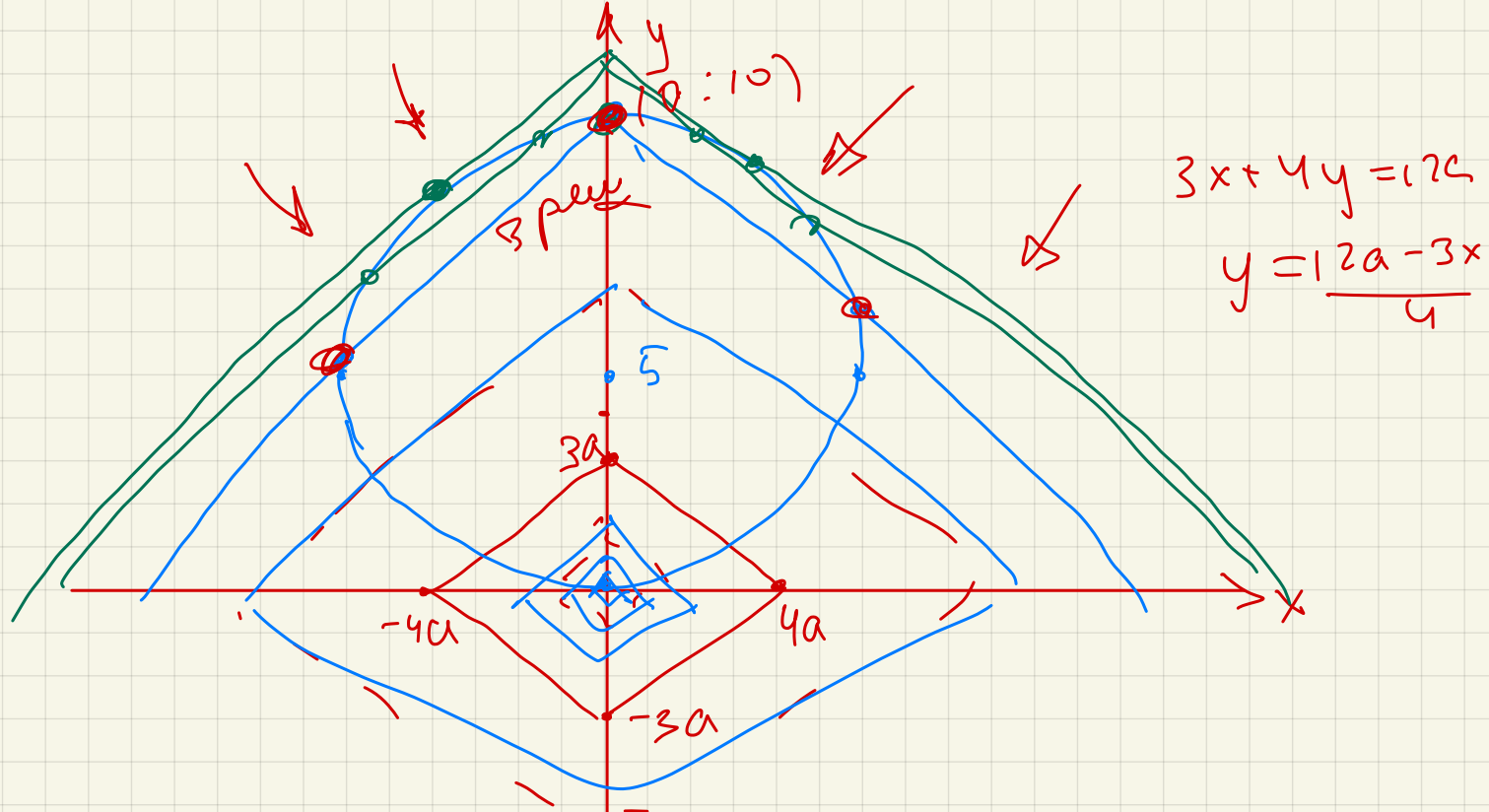
$$(0; 3)$$

$$(4; 0)$$

$$\textcircled{2} \quad -3x + 4y = 12$$

$$y = \frac{12 + 3x}{4}$$

$$(0; 3) \quad (0; -4)$$



$$x^2 + \left(\frac{12a - 3x}{4} - 5 \right)^2 = 25$$

$$x^2 + \left(\frac{12a - 3x - 20}{16} \right)^2 = 25$$

$$(12a - 3x - 20)(12a - 3x - 20) =$$

$$144a^2 - 36xa - \underline{\underline{240a}} - 36ax + 9x^2 - 60x - \underline{\underline{240a}} - 60x + 400$$

$$9x^2 - 72ax - 480a - 120x + 144a^2 + 400$$

$$16x^2 + 9x^2 - 72ax - 120x + 144a^2 + 480a + 400$$

$$= 406$$

$$25x^2 - x(72a - 120) + 144a^2 - 480a = 0$$

$$-4x(18a - 30)$$

$$-24x(3a - 5) + 144a^2 - 480a = 0$$

$$0 = (24)^2 (3a - 5)^2 - 4 \cdot 25 \cdot (144a^2 - 480a) = 0$$

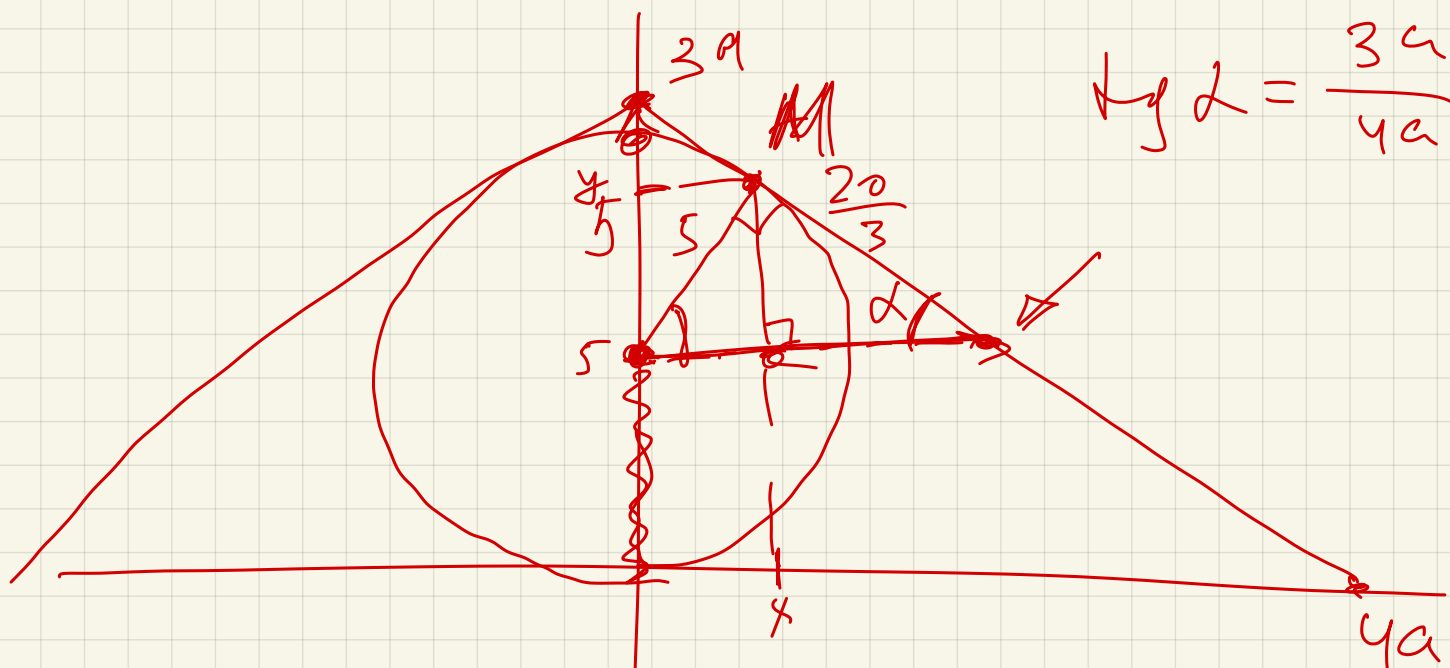
$$24^2 (9a^2 - 30a + 25) - 4 \cdot 25 \cdot 6 (3a^2 - 10a) = 0$$

$$24 \cdot 9a^2 - 24 \cdot 30a + 24 \cdot 25 - 25 \cdot 8 \cdot 3a^2 +$$

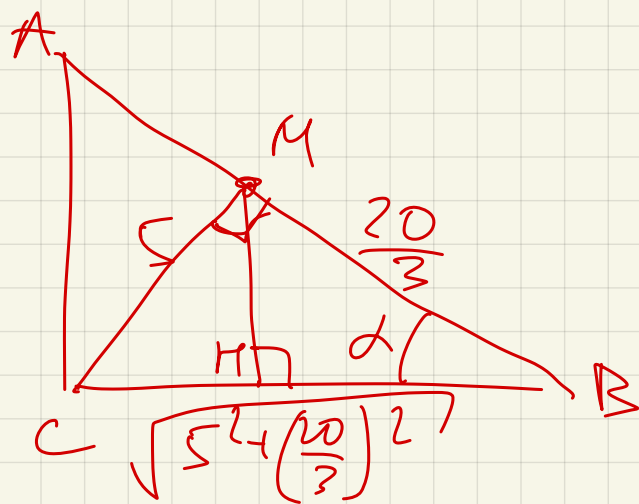
$$25 \cdot 8 \cdot 10a = 0$$

$$216a^2 - 600a^2 - 720a + 2000a + 600 - 384a^2 + 1280a + 600 = 0$$

0/3



$$\tan \alpha = \frac{3a}{4a} = \frac{3}{4}$$



$$\frac{CM}{CB} = \tan \alpha = \frac{3}{4}$$

$$CB = \frac{4 \cdot 5}{3} = \frac{20}{3}$$

$$CB \cdot \cos \alpha = CH$$

$$CM = CB - BM$$

$$CM = \frac{20}{3} \cdot \sin \alpha$$

$$\frac{20}{3}$$

$M(CH; 5 + MM) \rightarrow$